

TASAR COCOON COUNTING AND QUALITY CHECKING MACHINE WITH SEGREGATION

1. Title of the Idea

Tasar Cocoon Counting and Quality Checking Machine with Segregation

2. Abstract

India is the second-largest producer of silk in the world and the only country producing all four major commercial varieties of silk, that are Mulberry, Tasar, Eri and Muga. Tasar silk is mainly produced in states such as Jharkhand, Odisha, Bihar, Chhattisgarh and West Bengal. Thousands of farmers depend on tasar cocoons for their livelihood.

One major problem faced by tasar cocoon producers is the difficulty of accurately counting and assessing the quality of cocoons during selling. Unlike many agricultural products, tasar cocoons are commonly traded on a per-cocoon basis. Manual counting is slow, tiring and prone to errors. Existing counting methods may also cause physical damage to the delicate cocoons. Quality is often judged manually by pressing the cocoon, which is subjective and can damage the product.

This project proposes a low-cost automated machine that combines three important functions such as counting, quality checking and segregation. A conveyor belt gently carries the cocoons through the machine. Sensors detect and count individual cocoons. A light-based system examines the cocoon without physically damaging it and helps classify its quality. The machine then separates cocoons according to predetermined size and quality conditions. An Arduino-based microcontroller processes the sensor data and controls the system. An LCD display provides information such as the total number of cocoons counted and the number passing or failing the quality check.

The proposed machine aims to save time, reduce counting errors and prevent unnecessary damage. With an estimated large-scale production cost of around ₹15,000, it could provide an affordable technological solution for small-scale farmers and local sericulture units.

3. Why This Project?

Tasar silk farmers depend on accurate cocoon counting and fair quality assessment for proper payment. Manual counting is slow and error-prone, while pressing cocoons can damage them. A low-cost automated machine can count, inspect and segregate cocoons without unnecessary handling, saving time, reducing losses and helping farmers receive fairer prices.

4. Previous Studies, Problem Analysis and Data Elements

India has a well-established sericulture sector, with tasar silk forming an important source of livelihood in several eastern and central Indian states. Traditional cocoon marketing depends heavily on physical inspection and counting.

The main problems identified are:

1. Manual counting: Counting hundreds or thousands of cocoons individually requires considerable time and concentration.

2. Human error: Miscounting can result in incorrect payment or the need for repeated counting.
3. Physical damage: Frequent handling or pressing can damage the cocoon and reduce its market value.
4. Subjective quality assessment: Manual inspection depends on the experience and judgement of the person evaluating the cocoon.
5. Lack of integrated low-cost equipment: Counting, quality assessment and segregation are generally treated as separate activities.
6. Difficulty for small farmers: Sophisticated industrial equipment may be too expensive or complicated for small-scale producers.

Important Data Elements

The proposed system will collect and process:

- Number of cocoons passing through the conveyor.
- Cocoon size.
- Sensor response indicating the presence of a cocoon.
- Light transmission/scattering response.
- Quality classification based on predefined parameters.
- Number of cocoons accepted or rejected.
- Number of cocoons placed in each segregation category.

5. Description

A. Methodology

The project is designed as a school-level technological innovation addressing a practical problem in the tasar silk sector. The problem was studied from the perspective of cocoon farmers, particularly those involved in manual counting and quality assessment.

The proposed system uses a conveyor-based arrangement, sensors, light-based inspection, an Arduino microcontroller and an automated segregation mechanism.

The project development may be carried out in stages:

1. Study of the tasar cocoon-selling process.
2. Identification of difficulties in manual counting and quality checking.
3. Design of the machine.
4. Development of the counting mechanism.
5. Development of the optical quality-checking mechanism.
6. Integration of the segregation system.
7. Testing using representative cocoon samples.
8. Analysis of accuracy and improvement of the system.

The prototype is intended initially for controlled school-level testing. Large-scale farmer-field validation would be undertaken as a future stage.

B. Data Collected

Data Proposed to Be Collected During Testing

Parameter	Measurement
Total number of cocoons	Sensor-based count
Counting errors	Difference between actual and machine count
Quality response	Optical sensor reading
Accepted cocoons	Number passing quality criteria
Rejected cocoons	Number failing quality criteria
Processing speed	Cocoons processed per unit time
Segregation accuracy	Correctly separated cocoons

C. Activities Taken Up

- Studied the importance of tasar silk and tasar cocoons.
- Identified the problems associated with manual counting.
- Studied the need for non-invasive quality assessment.
- Designed a conveyor-belt-based counting system.
- Selected sensors for detecting individual cocoons.
- Proposed the use of light penetration/scattering for quality assessment.
- Designed an Arduino-based control system.
- Proposed automatic segregation according to size and quality.
- Designed an LCD display for real-time information.
- Considered affordability and energy efficiency during the design.

6. Proposed System and Working Principle

The machine consists of the following major sections:

1. Conveyor Belt System

A slow-moving conveyor belt gently transports the cocoons through the machine. This reduces unnecessary handling and helps prevent physical damage.

2. Automated Counting System

IR or other suitable sensors detect each cocoon as it passes through a defined detection zone. The sensor signal is sent to the Arduino microcontroller, which increases the cocoon count.

3. Optical Quality Checking

A controlled light source is directed toward the cocoon and the corresponding sensor measures the transmitted or scattered light. Differences in the optical response can be used to identify variations in cocoon condition. The process is non-invasive because the cocoon does not need to be pressed or opened.

4. Size and Quality Segregation

Based on the programmed conditions, a mechanical mechanism directs cocoons into different collection compartments, such as:

- Good-quality cocoons.
- Lower-quality cocoons.
- Different size categories.

The exact classification limits can be calibrated using actual cocoon samples.

5. Arduino Control

The Arduino receives information from the sensors, processes the readings and controls the conveyor, counting system, display and segregation mechanism.

6. LCD Display

The display can show:

- Total cocoons counted.
- Number passing the quality check.
- Number failing the quality check.
- Segregation status.

7. Scientific Principles Used

- Optical Physics
- Light penetration, transmission and scattering are used to obtain information about the internal characteristics of the cocoon without physically damaging it.
- Mechanical Engineering
- The conveyor mechanism provides controlled and gentle movement of cocoons through the different stages.
- Automation and Robotics
- Sensors, motors and actuators work together automatically to reduce manual intervention.
- Machine Vision / Infrared Detection
- IR or optical sensors can detect the presence and movement of individual cocoons for accurate counting.
- Microcontroller-Based Control
- An Arduino processes sensor information in real time and makes decisions according to programmed conditions.
- Data Analysis
- Sensor readings are interpreted to classify cocoons and control the segregation mechanism.
- Energy Efficiency
- Low-power electronic components can reduce operating costs and make the machine more suitable for rural use.

PART – II

9. Analysis of Data

The proposed machine can be evaluated by comparing its results with manual counting and visual quality assessment.

Counting Accuracy

For a test batch, the actual number of cocoons can first be determined manually. The same batch can then be passed through the machine.

$$\text{Counting Accuracy (\%)} = (\text{Number correctly counted} \div \text{Actual number of cocoons}) \times 100$$

This will indicate whether the sensor arrangement can reliably detect individual cocoons.

Quality-Checking Accuracy

Cocoons can be independently classified by experienced persons and compared with the machine's optical classification. This will help establish suitable sensor thresholds.

Segregation Accuracy

After processing, the number of cocoons correctly placed in each category can be recorded.

$$\text{Segregation Accuracy (\%)} = (\text{Number correctly segregated} \div \text{Total number tested}) \times 100$$

Time Saving

The time required for manual counting can be compared with the time required by the machine. A reduction in processing time would demonstrate one of the major advantages of automation.

Expected Findings

The proposed system is expected to:

- Reduce manual counting effort.
- Reduce counting errors.
- Reduce unnecessary physical handling.
- Provide a more consistent quality assessment.
- Speed up the selling and sorting process.
- Provide useful numerical information to farmers.
- Reduce dependence on multiple manual processes.

Because the quality-checking system requires calibration against real tasar cocoons, its final accuracy should be established through extensive testing before commercial use.

10. Suggested System / Material for Abatement of the Problem

The proposed solution is a low-cost Tasar Cocoon Counting, Quality Checking and Segregation Machine.

Main Components

- Conveyor belt.
- DC geared motor.
- IR/optical sensors.
- Light source for optical inspection.
- Light sensor/photodetector.
- Arduino microcontroller.
- LCD display.
- Motor driver.
- Servo motor or suitable actuator for segregation.
- Power supply.
- Frame and collection trays.
- Appropriate electronic wiring and protective casing.

The estimated cost of the machine, when developed on a larger scale, is approximately ₹15,000, subject to component prices and final design.

The machine is intended to bridge the gap between simple manual methods and expensive industrial automation.

11. Advantages and Impact

Accurate Counting- Automated counting can reduce errors associated with fatigue, distraction and repeated manual counting.

Quality Assessment- Optical inspection can reduce the need to press or physically manipulate each cocoon during quality checking.

Fair Pricing- Reliable counting and more objective quality classification can provide farmers with better evidence when negotiating the value of their produce.

Time Efficiency- A continuous conveyor system can process a large number of cocoons more quickly than manual counting.

Integrated Operation- Counting, quality checking and segregation are combined in one machine, reducing the need for separate equipment and repeated handling.

Farmer-Friendly Design- An LCD display and simple controls can make the system easy to operate with limited technical training.

Affordable Technology- The proposed cost is considerably more suitable for small-scale farmers than sophisticated industrial automation systems.

Wider Social Impact- If successfully developed and validated, the machine could improve transparency in cocoon marketing, reduce wastage and support the livelihood of tasar silk farmers.

12. Conclusion

The Tasar Cocoon Counting and Quality Checking Machine with Segregation is a proposed low-cost technological solution to an important problem faced by tasar cocoon farmers. Manual counting is time-consuming and prone to errors, while conventional quality assessment can be subjective and may damage the cocoon.

The proposed machine combines a conveyor belt, optical sensors, an Arduino-based control system, LCD display and automated segregation mechanism. It aims to perform three important operations—counting, quality checking and segregation—in a single workflow.

The innovation is particularly valuable because it focuses not only on automation but also on affordability and ease of use. By reducing manual handling and providing measurable information about the produce, the machine could help farmers save time, reduce losses and negotiate more confidently for fair prices.

Further testing with real tasar cocoons is necessary to calibrate the optical system, establish accurate quality parameters and determine the reliability of segregation. With improvements such as machine learning, IoT connectivity, solar power and additional sensors, the system could eventually be adapted for larger sericulture operations.

Thus, the project has the potential to connect traditional sericulture practices with affordable modern technology and contribute to a more efficient, transparent and farmer-friendly tasar silk industry.

13. Future Scope and Improvements

The project can be further developed in the following ways:

1. **Machine Learning:** A trained machine-learning model could identify defects and quality variations more accurately than simple threshold-based systems.
2. **IoT Connectivity:** Farmers could monitor cocoon counts, quality results and machine performance through a mobile phone.
3. **Solar Power:** Solar energy could make the machine more suitable for rural areas with unreliable electricity.
4. **Moisture Detection:** Additional sensors could measure cocoon moisture.
5. **Strength Assessment:** Suitable non-destructive methods could be investigated for estimating cocoon strength.
6. **Improved Machine Vision:** Cameras could identify size, colour, shape and visible defects.
7. **Industrial Scaling:** The conveyor and processing capacity could be increased for larger sericulture farms.
8. **Data Storage:** Records of batches, counts and quality classifications could be stored digitally.
9. **Better Calibration:** Testing with large numbers of real tasar cocoons could improve the reliability of optical quality assessment.
10. **Energy Optimization:** Low-power electronics and efficient motors could reduce operating costs.

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15. References Used

1. Central Silk Board, Ministry of Textiles, Government of India – Information and publications relating to sericulture, tasar silk and silk production.
2. ICAR – Central Tasar Research and Training Institute (CTRITI) – Information relating to tasar sericulture, tasar cocoons and cocoon production.
3. School-level reference material on Physics – Optics, particularly light transmission, reflection and scattering.
4. School-level reference material on Mechanical Engineering and Automation, particularly conveyor systems and sensors.
5. Arduino technical documentation and educational resources relating to microcontroller-based sensor systems.
6. Information gathered through discussions and observations concerning the traditional counting, inspection and selling of tasar cocoons.

Overall Project Statement

The proposed Tasar Cocoon Counting and Quality Checking Machine with Segregation is an integrated, low-cost and farmer-friendly system that combines automated counting, non-invasive quality assessment and segregation. Its ultimate objective is to reduce manual effort, prevent unnecessary cocoon damage, improve transparency and help tasar farmers obtain fair value for their produce.